Parser Implementation

I implemented the parser referencing the model from *Crafting Interpreters*.

Expression Grammar

Expression → Literal | Binary | Unary | Grouping

Literal → NUMBER | STRING | “true” | “false” | “null”

Binary → expression operator expression

Unary → (“-“ | “!”) expression

Grouping → “(“expression”)”

Operator → "==" | "!=" | "<" | "<=" | ">" | ">=" | "+" | "-" | "*" | "/" | “%”

No expression grammar changes were made since Lab 2.

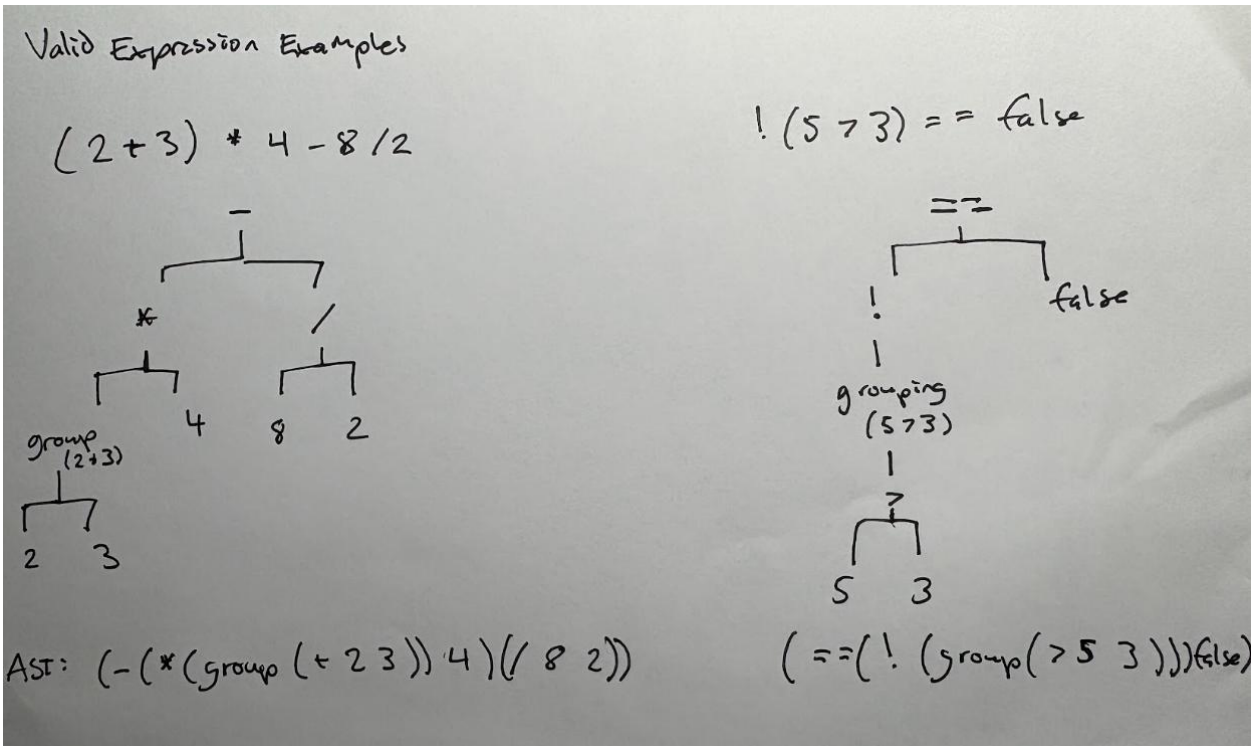
Precedence and Associativity

The precedence of the CN language follows that of lox:

Highest Precedence – Literals, Parentheses, Primary Expressions	(Primary Function)
Unary Operators	(Unary Function)
Multiplication, Division, Modulo Operators	(Factor Function)
Addition and Subtraction	(Term Function)
>, >=, <, <=	(Comparison Function)
==, !=	(Equality Function)

The CN language also follows lox in its left associativity.

Valid Expression Examples



The expression (2+3)*4-8/2 demonstrates the operator precedence in the CN parser. The parentheses force (2+3) to be evaluated before the multiplication operator. Multiplication and division have a higher precedence than subtraction and so 8/2 becomes the right operand of the tree. The AST is printed by the program, representing the expressions structure of the tree.

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC1-Valid_Expression1.py
(- (* (group (+ 2.0 3.0)) 4.0) (/ 8.0 2.0))
```

The same is true of !(5 > 3) == false. This expression demonstrates the precedence of unary, comparison and equality grammar. The equality operator has lower precedence than the grouping or literal and so the equality operator is made the root.

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC2-Valid_Expression2.py
(== (! (group (> 5.0 3.0))) False)
```

Extra Credit Opportunities

I added in the detection of missing left parentheses.

Testing

TC1-Valid_Expression1

Purpose

- Test the valid expression used to demonstrate the expression grammar, AST, and parser.

Expected Output: (-(*(group(+2.0 3.0))4.0/(/ 8.0 2.0))

Actual Output: Matched Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC1-Valid_Expression1.py
(- (* (group (+ 2.0 3.0)) 4.0) (/ 8.0 2.0))
```

TC2-Valid_Expression2

Purpose

- Test the second valid expression used to demonstrate the expression grammar, AST, and parser

Expected Output: (==(!(group(> 5.0 3.0)))false)

Actual Output: Matched Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC2-Valid_Expression2.py
(== (! (group (> 5.0 3.0))) False)
```

Test Cases 3-8

Purpose

- Test the literals: numbers with and without decimals, strings, true, false, null

TC3-Literal_Num

Expected Output: 15.0

Actual Output: Matched Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC3-Literal_Num.py
15.0
```

TC4-Literal_Decimal

Expected Output: 3.19

Actual Output: Matched Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC4-Literal_Decimal.py
3.19
```

TC5-Literal_String

Expected Output: hello world

Actual Output: Matched Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC5-Literal_String.py
hello world
```

TC6-Literal_True

Expected Output: True

Actual Output: Matched Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC6-Literal_True.py
True
```

TC7-Literal_False

Expected Output: False

Actual Output: Matched Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC7-Literal_False.py
False
```

TC8-Literal_Null

Expected Output: null

Actual Output: Matched Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC8-Literal_Null.py
null
```

Test Cases 9 -13

Purpose

- Test each arithmetic operator

TC9-Operator_Addition

Expected Output: (+ 5.0 9.0)

Actual Output: Matched Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC9-Operator_Addition.py
(+ 5.0 9.0)
```

TC10-Operator_Subtraction

Expected Output: (- 8.0 1.0)

Actual Output: Matched Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC10-Operator_Subtraction.py
(- 8.0 1.0)
```

TC11-Operator_Multiplication

Expected Output: (* 6.0 4.0)

Actual Output: Matched Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC11-Operator_Multiplication.py
(* 6.0 4.0)
```

TC12-Operator_Division

Expected Output: (/ 10.0 2.0)

Actual Output: Matched Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC12-Operator_Division.py
(/ 10.0 2.0)
```

TC13-Operator_Modulo

Expected Output: (% 8.0 4.0)

Actual Output: Matched Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC13-Operator_Modulo.py
(% 8.0 4.0)
```

Test Cases 14 -17

Purpose

- Test comparison and equality
- Test Precedence
- Test Associativity

TC14-Precedence1

Expected Output: (+ 2.0 (*3.0 4.0))

Actual Output: Matches Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC14-Precedence1.py
(+ 2.0 (* 3.0 4.0))
```

TC15-Precdence2

Expected Output: (==(< 2.0 3.0) True)

Actual Output: Matches Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC15-Precedence2.py
(== (< 2.0 3.0) True)
```

TC16-Associativty1

Expected Output: (/ (/ 20.0 5.0) 2.0)

Actual Output: Matches Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC16-Associativity1.py
(/ (/ 20.0 5.0) 2.0)
```

TC17-Associativty2

Expected Output: (- (- 10.0 3.0) (/ (* 2.0 4.0) 2.0))

Actual Output: Matches Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC17-Associativity2.py
(- (- 10.0 3.0) (/ (* 2.0 4.0) 2.0))
```

Test Cases 18 – 23

Purpose

- Test missing operand, unmatched parentheses, unexpected tokens errors

TC18 -Missing_Operand

Expected Output: Error at end: Expect Expression

Actual Output: Matches Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC18-Missing_Operand.py
[line 1] Error at end: Expect expression.
```

TC19 – Missing_Operand2

Expected Output: Error at '*': Missing left operand for *

Actual Output: Matches Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC19-Missing_Operand2.py
[line 1] Error at '*': Missing left operand for *.
```

TC20 -Unmatched_Paran

Expected Output: Error at end: Expected ‘)’ after expression.

Actual Output: Matches Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC20-Unmatched_Paran.py
[line 1] Error at end: Expect ')' after expression.
```

TC21 -Unmatched_Paran2

Expected Output: Error at “)”: Unexpected ‘)’ after expression.

Actual Output: Matches Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC21-Unmatched_Paran2.py
[line 1] Error at ')': Unexpected ')' after expression.
```

TC22 -Unexpected_Tokens

Expected Output: Error at “6”: Unexpected token after expression.

Actual Output: Matches Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC22-Unexpected_Token.py
[line 1] Error at '6': Unexpected token after expression.
```

TC23- Unexpcted_Tokens2

Expected Output: Error at character “@”: Unexpected character

Actual Output: Matches Expected Output

```
PS C:\Users\HP\Desktop\CNeumann_CIS343\CNeumann_CIS343\src> python cn.py TC23-Unexpected_Token2.py
[line 1] Error at character: @: Unexpected character
[line 1] Error at '9': Unexpected token after expression.
```